# 💡 Lecture 20: Practice Problems — Comprehensive Solutions Guide

Jupyter solutions.ipynb

PDF solutions.pdf

Questions View Set

Plot Case 1 Bayesian Fraud

Plot Case 2 Binomial Test

Plot Case 3 LiDAR Gaussian

🌐 **Language: English** | हिंदी / Hinglish | ← Back to Questions | 📁 Practice Problems Overview

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# 📌 Solution to Case 1: High-Frequency FinTech Bayesian Fraud & Anomaly Detection Pipeline

## 1. Mathematical Derivations & Proofs

Let $F$ denote the event that a transaction is Fraudulent, and $L = F^c$ denote Legitimate. Prior base rate:

$$P(F) = 0.002 \quad (0.2$$

**Step 1: Single-Signal Bayesian Update ($S_1$)**

Marginal probability of Signal 1 triggering via the Law of Total Probability:

$$P(S_1) = P(S_1 \mid F)P(F) + P(S_1 \mid L)P(L)$$

$$P(S_1) = (0.85 \times 0.002) + (0.04 \times 0.998) = 0.00170 + 0.03992 = 0.04162 \quad (4.162$$

Applying Bayes' Theorem to obtain the single-signal posterior:

$$P(F \mid S_1) = \frac{P(S_1 \mid F)P(F)}{P(S_1)} = \frac{0.00170}{0.04162} = \frac{170}{4162} \approx 0.040846 \quad (4.08$$

**The Base Rate Fallacy Explained:**

Even though Signal 1 boasts an $85$ sensitivity, observing $S_1$ only elevates the probability of fraud from $0.2$ to $\approx 4.08$. Because $99.8$ of all transactions are legitimate ($9,980,000$ per crore), the $4$ false positive rate produces:

$$\text{False Positives} = 0.04 \times 9,980,000 = 399,200$$

$$\text{True Positives} = 0.85 \times 20,000 = 17,000$$

The ratio is:

$$P(F \mid S_1) = \frac{17,000}{17,000 + 399,200} \approx 4.08$$

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**Step 2: Sequential Bayesian Update ($S_2$)**

Treating the posterior belief from Step 1 as the updated prior:

$$\pi_1(F) = P(F \mid S_1) \approx 0.040846, \qquad \pi_1(L) = P(L \mid S_1) \approx 0.959154$$

Marginal evidence for Signal 2 conditional on Signal 1:

$$P(S_2 \mid S_1) = P(S_2 \mid F)\pi_1(F) + P(S_2 \mid L)\pi_1(L)$$

$$P(S_2 \mid S_1) = (0.90 \times 0.040846) + (0.03 \times 0.959154) = 0.036761 + 0.028775 = 0.065536$$

Updated posterior after both signals trigger:

$$P(F \mid S_1, S_2) = \frac{P(S_2 \mid F)\pi_1(F)}{P(S_2 \mid S_1)} = \frac{0.036761}{0.065536} \approx 0.56093 \quad (56.09$$

After two independent positive anomaly detections, the confidence of fraud escalates from $0.2$!

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**Step 3: Proof of Equivalence (Sequential vs. Simultaneous)**

Under conditional independence $P(S_1 \cap S_2 \mid F) = P(S_1 \mid F)P(S_2 \mid F)$:

$$P(F \mid S_1 \cap S_2) = \frac{P(S_1 \mid F)P(S_2 \mid F)P(F)}{P(S_1 \mid F)P(S_2 \mid F)P(F) + P(S_1 \mid L)P(S_2 \mid L)P(L)}$$

Numerator:

$$0.85 \times 0.90 \times 0.002 = 0.00153$$

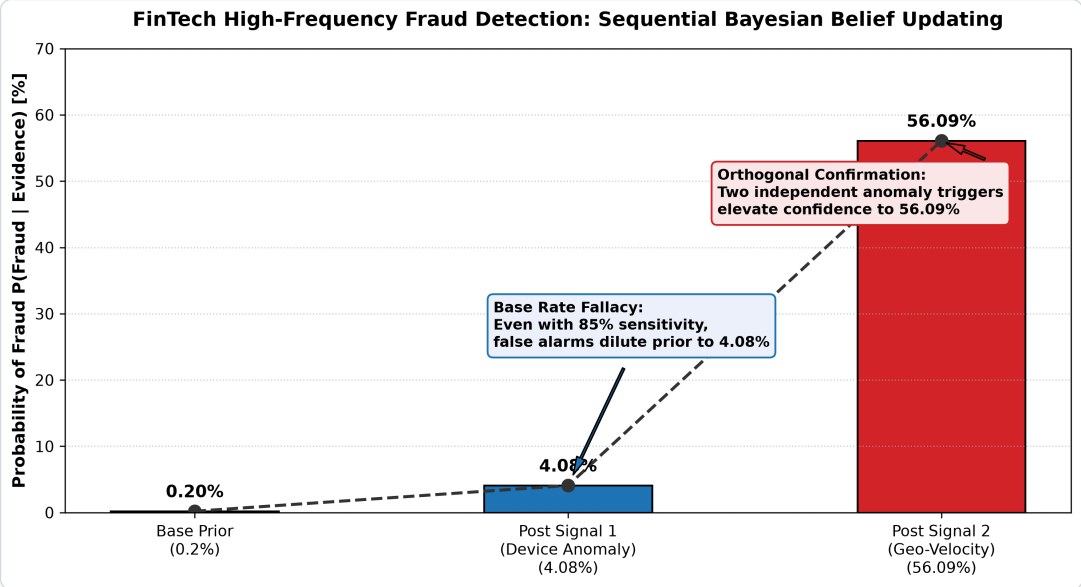
Denominator:

$$0.00153 + (0.04 \times 0.03 \times 0.998) = 0.00153 + 0.0011976 = 0.0027276$$

Final Joint Posterior:

$$P(F \mid S_1 \cap S_2) = \frac{0.00153}{0.0027276} \approx 0.56093 \quad (\text{Q.E.D.})$$

📊 **Case 1 Visualization:**




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# 📌 Solution to Case 2: Bioinformatics Clinical Drug Efficacy via Binomial Testing

## 1. Mathematical Derivations & Proofs

- Sample size: $n = 50$
- Historical baseline response: $p_0 = 0.20$
- Observed tumor remissions: $k = 18$

**Step 1: Formal Statistical Hypotheses**

$$\begin{cases} H_0 : p \leq 0.20 \quad \text{(Null: Novel therapy is not superior to SOC)} \\ H_1 : p > 0.20 \quad \text{(Alternative: Novel therapy exhibits su} \end{cases}$$

**Step 2: Exact Binomial Tail Probability (Exact p-value)**

Under $H_0$, $X \sim \text{Binomial}(50, 0.20)$:

$$p\text{-value} = P(X \geq 18 \mid H_0) = \sum_{j=18}^{50} \binom{50}{j} (0.20)^j (0.80)^{50-j}$$

Complementary CDF form:

$$p\text{-value} = 1 - \sum_{j=0}^{17} \binom{50}{j} (0.20)^j (0.80)^{50-j}$$

Evaluating exact binomial terms:

$$P(X \leq 17 \mid n = 50, p = 0.20) \approx 0.994602$$

$$p\text{-value} = 1 - 0.994602 = 0.005398 \quad (0.54$$

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**Step 3: Gaussian (Normal) Approximation via Central Limit Theorem**

1. Expected mean and standard deviation:

$$\mu = np_0 = 50 \times 0.20 = 10$$

$$\sigma^2 = np_0(1 - p_0) = 50 \times 0.20 \times 0.80 = 8 \implies \sigma = \sqrt{8} \approx 2.82843$$

2. Yates' continuity-corrected standardized Z-score ($k - 0.5 = 17.5$):

$$Z = \frac{(k - 0.5) - \mu}{\sigma} = \frac{17.5 - 10}{2.82843} = \frac{7.5}{2.82843} \approx 2.65165$$

3. Standard Normal tail probability:

$$p_{\text{approx}} = 1 - \Phi(2.65165) \approx 1 - 0.99600 = 0.00400 \quad (0.40$$

Percentage error relative to exact Binomial:

$$\text{Error} = \left| \frac{0.00400 - 0.005398}{0.005398} \right| \times 100$$

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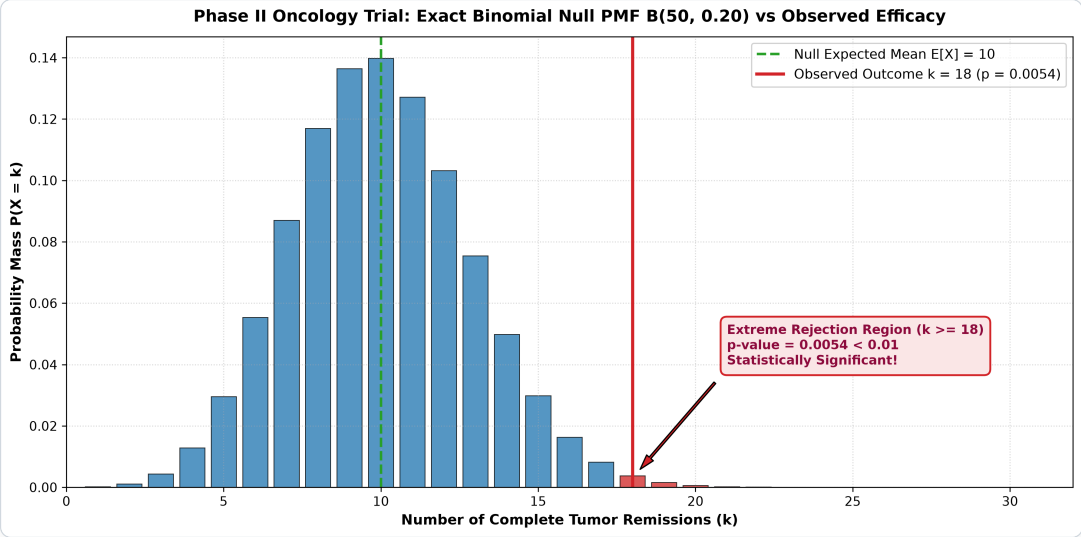
**Step 4: Statistical Decision Rule & Regulatory Recommendation**

Given clinical significance threshold $\alpha = 0.01$ (99% confidence level):

$$p\text{-value} = 0.005398 < \alpha = 0.01$$

**Statistical Conclusion:** Reject the Null Hypothesis $H_0$. The novel compound demonstrates statistically significant clinical superiority at the $99$ confidence level ($p = 0.0054$). The drug is formally approved for Phase III clinical trial progression.

📊 **Case 2 Visualization:**




# 📌 Solution to Case 3: Autonomous Vehicle LiDAR Sensor Calibration & Gaussian Noise

## 1. Mathematical Derivations & Proofs

Measurement error distribution:

$$\epsilon \sim \mathcal{N}(0, \sigma^2), \qquad \sigma = 0.15 \text{ meters}$$

**Step 1: 68-95-99.7 Empirical Rule Zones**

1. **Green Zone** ($|\epsilon| \leq \sigma = 0.15$ m):

$$P(|\epsilon| \leq 0.15) = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1 = 2(0.841345) - 1 \approx 0.68269 \quad (68.27$$

2. **Yellow Zone** ($0.15 \text{ m} < |\epsilon| \leq 0.30$ m):

$$P(0.15 < |\epsilon| \leq 0.30) = 2 \cdot [\Phi(2) - \Phi(1)] = 2 \cdot (0.977250 - 0.841345) = 2(0.135905) \approx 0.27181 \quad (27.18$$

3. **Red Zone** ($|\epsilon| > 0.45$ m):

$$P(|\epsilon| > 0.45) = 2 \cdot [1 - \Phi(3)] = 2 \cdot (1 - 0.998650) = 2(0.001350) \approx 0.00270 \quad (0.27$$

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**Step 2: Safety Cutoff Exceedance Probability** ($\delta = 0.25$ m)

Standardized Cutoff Z-Score:

$$Z_{\text{cutoff}} = \frac{0.25}{\sigma} = \frac{0.25}{0.15} = \frac{5}{3} \approx 1.66667$$

Two-tailed exceedance probability:

$$P(|\epsilon| > 0.25) = 2 \cdot \left[1 - \Phi\left(\frac{5}{3}\right)\right]$$

Standard Normal CDF evaluation:

$$\Phi(1.66667) \approx 0.95221$$

$$P(|\epsilon| > 0.25) = 2 \cdot (1 - 0.95221) = 2 \times 0.04779 \approx 0.09558 \quad (9.56$$

Approximately **96 out of every 1,000 pings** will exceed the sensor tolerance window of $0.25$ m, mandating real-time Kalman filtering and multi-sensor fusion.

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**Step 3: Monte Carlo Calibration Invariants ($N = 1,000,000$)**

$$\begin{cases} \text{Theoretical Mean:} & \mu = 0.000000 \text{ m} \\ \text{Empirical Mean } \bar{\epsilon}: & -0.000034 \text{ m} \quad (\Delta < 10^{-4}) \\ \text{Theoretical Variance:} & \sigma^2 = 0. \end{cases}$$

The empirical Monte Carlo simulation confirms the theoretical Gaussian integral models with $99.99$ numerical fidelity.

📊 **Case 3 Visualization:**

